

Measuring the Response of a Typical Brazilian Masonry Type Structure to Coal Mine

Blasting Vibrations

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Abstract

The present study was conducted in an open pit coal mine in southern Brazil where blasting is conducted on a daily basis as close as 100 m from residential structures. The objective of this study was to better understand the effects of ground vibrations and air sound pressure levels on structure movements, and specially, on the potential for wall damaging. A typical masonry built structure, commonly built in Brazil, was instrumented with velocity sensors to measure whole structure and mid-wall motions in order to compute blast-induced global wall strains and estimate the potential for causing damages to the structure walls, such as superficial cracks. Displacement gages were mounted over an existing wall crack to measure changes in crack width resulting from daily and long-term variations in weather conditions and compare to blast-induced crack response. The results show that strains resultant from blasting activities were lower than the failure strains for the weakest material composing the structure walls and long-term weather variations show higher influence over the instrumented crack compared to the blasting vibrations impacts.

Keywords: coal blasting; blasting vibrations; structure response to blasting; blast-induced wall strain; damages from blasting; crack potential.

Introduction

Coal has been mined in southern Brazil since late 1800's and in the beginning all mines were underground, which allowed the development of residential communities on the surface areas near the mine shafts. Nowadays, the same deposits have been mined open pit and with the expansion of the communities into towns, residential areas are very common close to the operations. Blasting becomes a major concern as it generates a potential nuisance to these nearby communities, resultant from ground vibrations and air sound pressure levels. There are many complaints about wall cracking and other structure damages associated with blasting activities, especially when blasting is conducted very close (between 100 and 300 m) to residential structures.

The objective of this study was to evaluate the impact of blast-induced ground vibrations and air sound pressure levels on structures by measuring the structure movements during these excitations and estimating the levels of strain induced in the walls. The resultant computed strains were compared to the failure strains of the materials composing the walls, and a damage potential could be determined. Other forces, such as weather fluctuations were also evaluated and compared to blasting effects. Such forces start to act from the moment the structure is built and continue working during its entire life (Oriard 1999). In the end, it was possible to determine if blasting can be considered the major contributor for structure damages.

Typical Brazilian structures are built with load bearing clay bricks and cement-mortar. Foundations comprise concrete beam perimeter walls and/or thin concrete slabs. Roofs are heavy as they are construction with clay tiles over a wood frame. Typically, walls are covered with cement grout, spackling (or plastered) and then painted. Interior walls are made of clay bricks and mortar, in the same manner as the exterior walls.

Measuring Structure Movements

Velocity transducers were mounted in the structure walls according to Fig. 1. Single-axis velocity transducers were mounted in the interior of the structure in the upper corner of the walls near

the ceiling (*S2*), in the lower corner, near the foundation (*SI*), and on the mid-walls (*MW*). A tri-axial geophone was buried in the ground in the exterior of the structure close to the instrumented corner (*GI*), to record the ground vibrations driving the structure motions. A microphone was placed in the same location to record air sound pressure levels (Aimone-Martin and Eltschlager 2003).

Ground vibrations are typically recorded in three directions, longitudinal (or radial, *R*), transverse (*T*) and vertical (*V*), and for this reason the geophone installed in the exterior of the structure was oriented in order the horizontal components (*R* and *T*) were aligned parallel with the structure axis, as shown in Fig. 1. Four channel blasting type seismographs, manufactured by White Seismology, Inc., were used to digitally record ground vibrations, air sound pressure levels and wall velocities during blast-induced wall movements. The three sensors in the upper corner, plus a mid-wall sensor, were connected to one seismograph (*S2*), while the three lower corner and the other mid-wall sensors were connected to another seismograph (*S1*). The exterior ground geophone and microphone were connected to a third seismograph (*GI*). All seismographs were connected in series and the structure seismographs (*S1* and *S2*) were configured in manual mode, with the exterior unit being responsible for triggering the system during blasting events and to send a signal to the other units to start recording data. With this configuration, all data recorded was time correlated.

Data was processed and analyzed using Seismograph Data AnalysisTM software, provided by the seismograph manufacturer. Peak particle velocities, peak frequencies, predominant frequencies (determined using FFT – Fast Fourier Transform analysis) and peak displacements for each vibration component, in the ground and structure walls, were obtained from the seismograph reports provided by the software for each event. Peak wall movements were correlated with peak ground vibrations and air sound pressure levels for each event recorded and the response of the structure was evaluated.

The Natural frequency of the structure was determined using a technique involving transfer functions calculated from Fourier frequency analysis (Dowding 1985). The method consists of applying the Fourier transform integral to the velocity time-history waveform to yield the power spectral density function. The ratio between the square of the Fourier amplitude of the upper structure

(S2) and ground (GV) was obtained and compared to the square of the spectral density of the upper structure and ground response to determine the maximum structural response. The resulting frequency of the maximum structural response was considered the natural frequency of the structure. Damping was calculated through the half power bandwidth method, where damping measurements are based on the frequency response (Cossolino and Pereira 2010).

Velocity measurements recorded by the seismographs were integrated in the Seismograph Data AnalysisTM software to determine actual wall displacements and the differential displacements between upper and lower corner movements were computed ($S2 - S1$). The maximum differential wall displacement (δ_{max}) was then used to determine the maximum induced shear strain (γ_{max}) in the wall (Dowding 2000). In-plane tensile strains (ϵ_{Lmax}) is the deformation mode most likely to cause cosmetic cracks in walls during high levels of structure motions and is directly related to global shear strains in the walls (Aimone-Martin et al. 2003). Mid-wall bending strains are directly related to bending stresses induced in the mid-walls (Dowding 2000). The geometry of the walls and equations used to compute global shear strains, in-plane tensile strains and bending strains are shown in Fig. 2.

Calculated strains were compared with the failure limits of the weakest material used in the construction, and the probability of causing superficial cracks in the walls was estimated (Aimone-Martin et al. 2003). The cement mortar used between bricks and as wall coating (interior and exterior walls) was considered the weakest material in the structure constitution and, therefore, the strain limits of this material were used to evaluate crack potential. Studies conducted in cement mortar indicates the failure limits of the cement grout by shear strain is approximately 600 micro-strains (10^{-6} mm/mm) and by in-plane tensile and mid-wall bending strains ranges between 150 and 250 micro-strains (Paes and Carasek 2002, Mohamad et al. 2009, Silva et al. 2009 and Bastos et al. 2010).

In order to measure the influence of weather fluctuations on the structure walls, an existing crack was monitored over a long period of time. To measure crack movements induced by long-term and daily variations in temperature and humidity, as well as blasting vibrations, a displacement gage

was mounted over an existing wall crack, as shown in Fig. 3. Crack width variations were correlated with ground vibrations and air sound pressure amplitudes (dynamic response). Aperture variations were measured on an hourly basis over the period of study and correlated with daily fluctuations of temperature and humidity (static response). A displacement gage was also installed on a portion of undamaged wall (null gage in Fig. 3.) to measure the natural expansion and contraction of the material, and this result was subtracted from the measurements of the crack aperture, in order to isolate crack movement (Dowding 2008).

Mine Geology and Blast Design

The coal deposit at the studied operation is located in the Butiá-Leão Coalfield, located in the Parana Intracratonic basin, in southern Brazil. The individual coal seams have thicknesses varying from 0.5 m up to 2.2 m, composed of levels of bright coal rich in vitrinite, alternating with levels of matte coal and carbonaceous siltstones. Density of the coal seams range from 1.5 to 1.8 t/m³ as a function of ash content and partings. Overburden varies from 5 meters to over 30 meters. Fig. 4. shows the typical profile of the coal deposit, where a striping mine operations is conducted and all seams are exploited selectively, including the remaining pillars from an old underground mine.

In order to mine the coal seams selectively, blasting must be conducted in a manner that mixture of coal and waste material is avoided. For this reason, low powder factors (amount of explosive per volume of blasted rock) are used in the blast design, resulting in tight and confined blasts with little rock movement. A minimum amount of energy is used in order to induce fractures to the coal and siltstones to facilitate the excavation. Powder factors range from 150 g/m³ in overburden benches to 270 g/m³ in coal benches. These types of blast can lead to high levels of ground vibrations due to the large drill patterns and confinement of the holes. Fig. 4. presents the blast benches and the top three are considered overburden blasts, while the bottom two are coal blasts. The average blast hole length was 5 m drilled in a 76 mm diameter, arranged in staggered patterns. Drill patterns were mainly 4 x 5 m in the overburden benches and 3 x 4 m in the coal benches.

Results

Over the six months of study, 148 blasts were conducted in the coal mine and 115 events were recorded by the instruments placed in structure. Fig. 5. shows the location of the structure near the blasting area and a photograph of the instrumented masonry built structure. The foundation of the house was seated on a soil developed on top of clay and siltstones that are part of the geological sequence. The structure distances from the blasts ranged from 33 m to 810 m, and blasts charge weights per delay varied between 13.8 kg to 245.9 kg, resulting in scaled distances of 3.5 to 125.6 m/kg^{1/2}. Scaled distance is a means of incorporating the two most important factors contributing to the intensity of ground motions produced by blasting, as intensity decreases proportionally with the distance and inversely with the explosive weight detonated in one-time delay. In the case of ground motion, the square root of the scaled distance is used (commonly referred to as simply scaled distance, SD), as ground motion has been shown to correlate best with the square root of the charge weight detonated (Dowding 1985). The scaled distance factor is defined by Eq. 1.

$$SD = \frac{D}{\sqrt{W}} \quad (1)$$

Where, D is the distance from the blast and W the charge weight per delay period. The scaled distance factor allows the comparison of blasts of different sizes (Siskind et al. 1980).

Peak particle velocities (PPV) recorded in the soil near the foundation of the monitored structure ranged from 1.27 mm/s to 83.31 mm/s, with peak frequencies ranging from 3.9 Hz to 36.5 Hz, averaging 15.5 Hz. The maximum peak particle velocity of 83.31 mm/s was recorded for a blast located 33 m from the structure, with a charge weight per delay of 58.3 kg. Fig. 6a. presents the attenuation plot where peak particle velocity was correlated with scaled distance and the data for each event recorded at the structure and in additional monitoring points were used to determine the attenuation equation for the operation. A power curve best-fit line through the data defines the attenuation equation and this equation has the form presented in Eq. 2.

$$PPV = k * SD^{-b} \quad (2)$$

Where, k is the y-intercept at SD equals to 1 and b is the slope of the attenuation line. The k factor measures the intensity of the seismic energy transferred into the ground and propagated away from the blast site, and it is a function of confinement, explosive density and energy. The slope term b is related to the geology through which the seismic wave passes and is a measure of the decay of the velocity intensities with distance (Dowding 1985).

The high y-intercept of 317.1 and the slope of 1.35, shown in the equation in Fig. 6a., indicate that the high levels of vibration measure close-in to the blasts are dissipated (attenuated) quickly with the increase in distance from the blast. The coefficient of determination (R^2) of 0.76 indicates the data is well correlated and PPV values predicted by this equation are very close to real measurements. Fig. 6b. shows the Brazilian standard limits for blasting in urban areas (ABNT NBR 9653 2005), which is based in peak particle velocity and peak frequency and as the frequency of ground vibrations increase, the allowed peak particle velocity also increases. The 115 blasts recorded at the structure were included in the graphic and it was observed that eleven (11) events presented peak particle velocities above the limits suggested by the Brazilian standard, but not apparent damages were observed in the structure after theses blasts.

Average ground peak frequency for all data recorded at the structure was 15.5 Hz, while the average predominant frequency, FFT, was 12.9 Hz. The overburden blasts, at benches SS, CA and SC, average peak frequency was 12.8 Hz, and FFT frequency was 8.9 Hz, and blasts in the coal seams generate seismic waves with propagation frequencies averaging 19.5 and FFT around 18.8 Hz. This indicates blasting in the overburden siltstone generated lower frequencies compared to blasting at the lower benches, at the coal seams. It was also observed for the overburden blasts that blasting near the structure, under 150 m resulted in average ground frequencies of 15.4 Hz, and average predominant frequencies of 11.9 Hz, while blasting at distances over 150 m, the resultant average frequencies were 11.6 Hz and 7.5 Hz, peak and predominant FFT frequencies, respectively. This shows that the further the distance from the structure, lowest the frequency in the ground, and this was associated to the fast

dissipation of the seismic energy through the ground, as shown by the attenuation equation on Fig. 6a. It was also believed the presence of the underground mine influences the propagation frequency of the blast vibrations, as it creates a reflection surface and somehow reduces the frequency of the waves. As the coal seams are at the same level and below the underground mine no reflection occurs and the seismic waves propagate with higher frequencies.

Structure Response Analysis

Fig. 7a. presents a schematic drawing in-plan view of the structure with the location of the instrumented corner and mid-walls, as well as the monitored crack. In Fig. 7b., photographs of the installed sensors in the walls and in the ground are shown. The sensors were placed so the south wall contained the longitudinal (radial) velocity transducer perpendicular to the wall plane. The transverse velocity transducer was mounted perpendicular to the east wall plane.

Fig. 8. show correlations between lower corner (S1) movement and peak ground vibrations (GV), in terms of peak velocity and peak displacement. Lower structure peak velocities, Fig. 8a., demonstrate a small decrease of ground vibrations for the horizontal components (R and T), indicated by the slope of the line below one (1), while the foundation movement in the vertical direction is higher than the ground excitation. Once peak velocities were integrated into peak displacements, Fig. 8b, the actual displacements in the vertical direction were smaller than in the horizontal components and lower than the ground displacements. This indicates that the foundation of the structure was well coupled to the ground, presenting a decrease in amplitudes when the seismic energy is transferred from the ground into the structure.

The correlation of upper structure movements and ground excitations was presented in Fig. 9. Amplification of the ground vibrations was observed in the upper structure, indicated by the slopes of 1.37, 1.29 and 1.15, in the radial, transverse and vertical directions, respectively, shown in Fig. 9a., where peak ground velocity was correlated with peak upper corner velocity. In terms of peak

displacement, Fig. 9b., the horizontal directions of movement present a small amplification of ground displacements, while in the vertical direction a decrease of ground amplitudes was observed. In both upper and lower corners, it was observed the vertical displacement of the structure was much smaller than the movement in the horizontal directions and, for this reason, it was assumed the horizontal components of ground vibration most likely would result in the highest induced wall strains.

Mid-walls movement were correlated with ground vibrations in Fig. 10. and it was observed the racking motion of the transverse ground component has more influence on the east mid-wall movement, compared to the direct hit of the longitudinal component in the south mid-wall. Fig. 10a. shows the correlation of peak mid-wall velocities with ground excitations, and the influence of the ground frequency in the structure mid-walls responses can be observed. When ground frequencies were low, mid-wall velocities were higher than when ground frequencies were higher. In terms of displacements, Fig. 10b., the influence of ground frequencies was not noticeable.

The frequency analyses resulted in a Natural Frequency of the structure ranging from 10.2 to 11.5 Hz, depending on the wall. Natural frequency was also determined for the mid-walls and were defined in 12.2 and 11.4 Hz for the south and east wall, respectively. Average damping ratio was calculated 1.24 %, indicating the seismic energy was dissipated very fast by the structure walls and no free-response, or oscillatory movement was observed in the structure.

In Fig. 11., air sound pressure levels are correlated with upper structure and mid-wall displacements and was observed the data gathered for the blasts in the overburden benches, closest to the surface, presented higher correlation with the highest levels of structure displacements, both in the upper structure and mid-walls. Sound pressure levels originated from coal benches, deep down in the mine pit, were low and did not appear to have much effect on structure movements. The influence of air sound pressure levels over the structure movements is lower compared to than the influence of ground vibrations, and correlations present high scatter, indicated by the lower determination coefficients (R^2). Despite the lower influence of sound pressure levels, it contributes to structure interior noise, as it happens to provoke windows to tremble and sometimes the ruff frame.

Table 1 shows the computed differential wall displacements, shear strains, in-plane tensile strains and mid-wall bending strains for the eleven events resulting in ground vibrations above the Brazilian standard limits for blasting in urban areas. The resultant differential displacements in the vertical direction were 6.8 to 7.8 times smaller the computed differential displacements for the horizontal components, and for this reason the strains in the vertical component were not considered in this analysis. The maximum computed differential wall displacement between upper and lower structure ($S_2 - S_1$) was 0.603 mm, computed for the blast at 33 m from the structure, resulting in a maximum shear strain (γ_{max}) of 241.2 micro-strains and in-plane tensile strain (ϵ_{Lmax}) of 71.6 micro-strains. Maximum bending strain (ϵ) was computed 130.5 micro-strains. It was observed that the highest mid-wall bending strain in the east wall was not resultant from the highest ground excitation of 48.77 mm/s in the transverse direction, but occurred during the excitation of 35.56 mm/s. This happens because for the blast resulting in the highest mid-wall bending strain, the whole structure movement was greater than for the blast with the highest ground velocity, and as show in Fig. 2c., the position of the whole wall needs to be considered in the mid-wall bending strain calculations, besides the recorded movement of the mid-wall.

These strain levels were compared to the failure strains for the cement grout used as mortar in between clay bricks and for wall coating, and a damage potential was determined. The cement grout was considered to be the weakest material in the wall construction and most likely to sustain cosmetic damages. The maximum computed in-plane tensile strain was 2.1 times smaller than the failure strain of 150 micro-strain defined for the cement grout, while the maximum computed bending strains was 1.1 times the failure strain of this material. The ration between the computed wall strains and failure strains of the cement grout (damage potential) the eleven blasts with highest ground vibrations were presented in Table 1.

In Fig. 12., correlations between in-plane tensile strains and bending strains with the driving peak particle velocities recorded in the ground for all events were presented. The induced in-plane tensile strains in both wall are similar, indicating ground vibrations in either directions result in similar

impact to the structure walls. For bending strains, the racking motion of the transverse direction shows greater influence in the bending of the east mid-wall than the direct hit of the radial ground velocities in the south wall.

Crack Response Analysis

Long-term changes in crack width along with variations in outside temperature and humidity for a period of 160 days (3851 hours) were plotted in Fig. 13. In general, long-term crack movements followed the trend in ambient humidity, when average daily humidity decreased, crack tended to close (grey arrows in Fig. 13.). Short-term (or 24 hours) movement was consistent with daily variations of temperature, when humidity increased, the crack opened (positive change) whereas a sudden increase in temperature resulted in crack closure (dashed rectangle in Fig. 13.).

During the winter and spring, weather fluctuations presented higher influence over the crack width variation, resulting in higher changes in aperture. Over the summer, when variations in temperature and humidity were more constant, crack movements were also stable (black arrows). The total long-term opening of the crack was computed to be 0.403 mm, while the maximum daily crack aperture was 0.181 mm at a maximum temperature variation of 15.1°C and maximum humidity change of 50.9%, occurring during the winter. The maximum daily temperature variation of 27.7°C occurred during the spring, and crack moved 0.133 mm with a humidity variation of 65.4%. Whereas, on the day of the maximum humidity variation of 69.2%, crack aperture changed 0.148 mm and temperature varied 26.1°C. This shows that crack opening and closing has similar responses to daily variations of both temperature and humidity and is more sensitive to humidity changes. During weather fronts, such as rain, crack movements were minimum, indicated in Fig. 13., and the crack width variation remains nearly constant with the small variation in temperature and humidity during raining days. The scatter plot shown in Fig. 14. presents daily variations (maximum – minimum, then minimum – maximum), or 12-hour cycles, in crack aperture (ΔC) correlated with variations in temperature (ΔT) and humidity (ΔH). This plot shows that when temperature increases (negative

variation), humidity decreases (positive variation) resulting in crack closure (negative variation), while with the increase in humidity (especially at nighttime), temperature falls and the crack opens (positive variation).

Blast-induced crack response was recorded for 70 out of the 115 events triggered by the ground sensor. Issues with the crack gage system prevented the recording of 45 blasts. Nonetheless, it was possible to evaluate crack response over a wide range of ground vibrations, between 1.27 to 28.96 mm/s, and air sound pressure levels from 2 to 380 dB. Blast-induced crack movements ranged from 0.0078 to 0.3087 mm (78 to 308.7 μm), peak-to-peak displacement. Fig. 15. shows correlations of maximum crack aperture with peak particle velocity and air sound pressure levels. Fig. 15a. indicates the horizontal movements of the ground have similar influence on crack movement, and it is possible to observe the influence of ground frequency on crack displacements. When ground frequencies were over 11 Hz (the natural frequency of the structure), crack changes were also elevated. Air sound pressure levels have little influence in crack movement, shown by the high scatter in the data presented in Fig. 15b. The maximum change in crack aperture during blast events was 0.309 mm, resultant from blasting 125 m from the structure. This dynamic response was 1.3 times smaller than the long-term weather-induced overall crack displacement of 0.403 mm.

Seven blasts, out of the 70 events evaluated, resulted in crack aperture greater than the maximum crack displacement from daily weather-induced variations, of 0.181 mm. Peak particle velocity of four of these events exceeded the Brazilian standard limits and all seven blasts were within 130 m from the structure. Even though these blasts resulted in crack movements greater than the maximum daily crack width variations cause by variations in temperature and humidity, no extension or widening of the existing cracks or the formation of new cracks was observed after these blasts.

Blasting in the studied Coal Mine was as close as 100 m from off-site structure, and the blasts conducted during the study period over this distance resulted in maximum in-plane tensile strains of 26.1 micro-strains and maximum bending strain of 26.9 micro-strains in the evaluated structure. These values are 5.7 and 9.3 times smaller than the failure limits for the cement grout. Therefore,

blasting practices applied by the mine company cannot possibly result in damages to the structures located near the operation.

Conclusions

The following conclusions can be drawn from this study:

- The masonry structure studied showed movements in the foundation and upper structure following very close the ground excitations resultant from blasting activities, and for this reason the structure did not present free response, or oscillatory movement after the passing of the seismic waves;
- The strain levels computed, induce by blasting events, were lower than the failure limits for the weakest material comprising the structure. The stresses and deformations induced in the structure walls by ground vibrations did not exceed the elastic limits of the materials and no permanent deformation was observed resultant from blasting activities. Therefore, cracks in the cement mortar and wall coating, in the exterior or the interior of the structure, cannot be associated to blasting vibrations;
- The maximum blast-induced crack width change was 1.3 times smaller than the overall weather-induced long-term crack aperture variation registered during the study. Seven blasts resulted in crack movements greater than the daily influence of temperature and humidity. However, no new cracks were observed after these blasting events;
- The effects of climate can be considered as the biggest factor contributing to changes in crack aperture over time. The daily variations of temperature and humidity over long periods of time can fatigue materials comprising the structure, resulting in superficial and cosmetic damages;
- Crack aperture variations were more sensitive to weather changes than to blast-induced motions. Residents living near blasting operations usually associate damages in the

residences to blasting vibrations, however, in many cases ground vibrations and air sound pressure levels effects over the materials that compose the structure walls are very low compared to the slow, long-term and almost imperceptive stimulus resultant from daily fluctuations in temperature and humidity. It is not likely that blasting is the source of structure damage when compared with the influence of weather.

- This work also shows that even though a few blast events resulted in peak particle velocity greater than the limits proposed by the Brazilian standards, these levels of vibration did not cause damages to the studied structure. This indicates the Brazilian standard limits for safe blasting in urban areas is conservative and peak particle velocities below these limits cannot possibly result damages to structures.

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389 **Fig. 1.** Position of single-axis velocity sensor used to measure structure response

390 **Fig. 2.** Geometry of the wall and equations used to compute shear strains (a) in-plane tensile strains

391 (b) and mid wall bending strains (c)

392 **Fig. 3.** Displacement gages mounted over an existing wall crack and in the undamaged wall

393 **Fig. 4.** Coal deposit profile

394 **Fig. 5.** Location map showing the location of the structure near the blasting area (a) and the

395 instrumented structure (b)

396 **Fig. 6.** Attenuation plot correlating peak particle velocity with scaled distance (a), peak particle

397 velocity versus peak frequency the Brazilian standard limits [14] (b)

398 **Fig. 7.** Schematic drawing in-plan view of the structure (a), photos of the instrumented walls and

399 ground geophone installation (b)

400 **Fig. 8.** Correlation of ground vibration (GV) with lower corner movement (S1), in terms of peak

401 velocity (a) and peak displacement (b)

402 **Fig. 9.** Correlation of ground vibration (GV) with upper corner movement (S2), in terms of peak

403 velocity (a) and peak displacement (b)

404 **Fig. 10.** Correlation of ground vibration (GV) with mid-wall movement (MW), in terms of peak

405 velocity (a) and peak displacement (b)

406 **Fig. 11.** Correlation of sound pressure level (SPL) with upper structure (a) and mid-wall (b)

407 displacements

408 **Fig. 12.** Peak ground velocity influence on in-plane tensile strains (a) and bending strains (b)

409 **Fig. 13.** Weather-induced daily changes in crack aperture

410 **Fig. 14.** Daily, or 12-hour cycles, variations in crack aperture correlated with variations in

411 temperature and humidity

412 **Fig. 15.** Crack response to peak ground velocity (a) and sound pressure levels (b)

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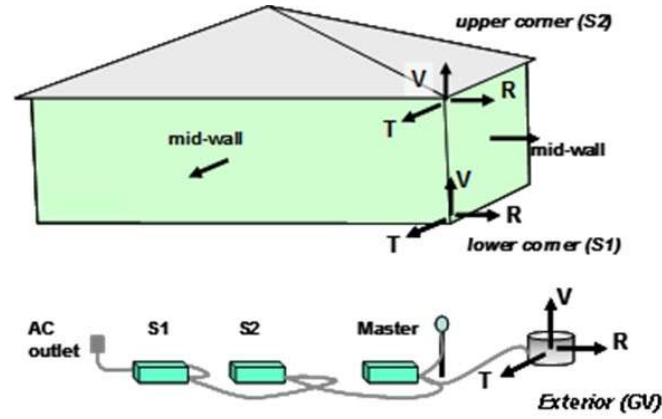


Fig. 1. Position of single-axis velocity sensor used to measure structure response

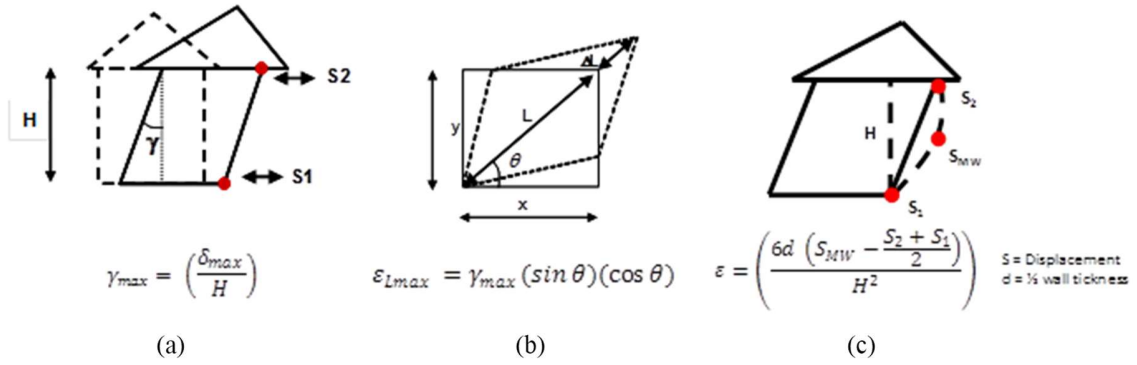


Fig. 2. Geometry of the wall and equations used to compute shear strains (a) in-plane tensile strains (b) and mid-wall bending strains (c)

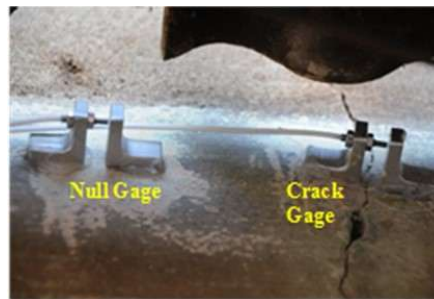


Fig. 3. Displacement gages mounted over an existing wall crack and in the undamaged wall

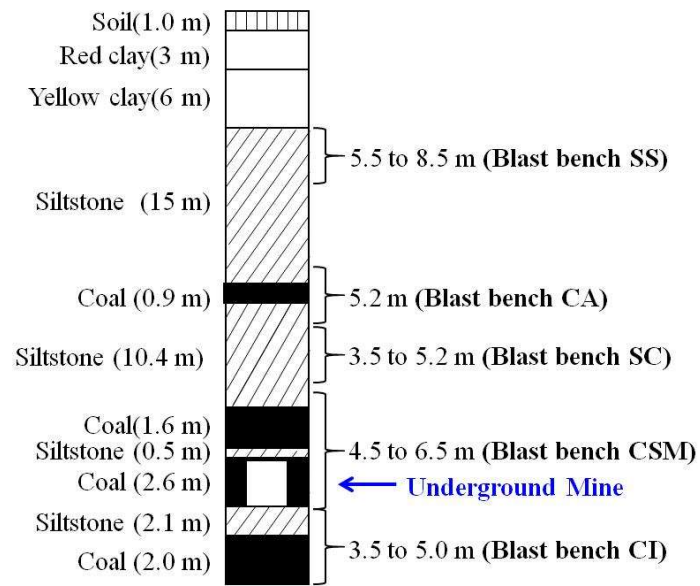


Fig. 4. Coal deposit profile



(a)



(b)

Fig. 5. Location map showing the location of the structure near the blasting area (a) and the instrumented structure (b)

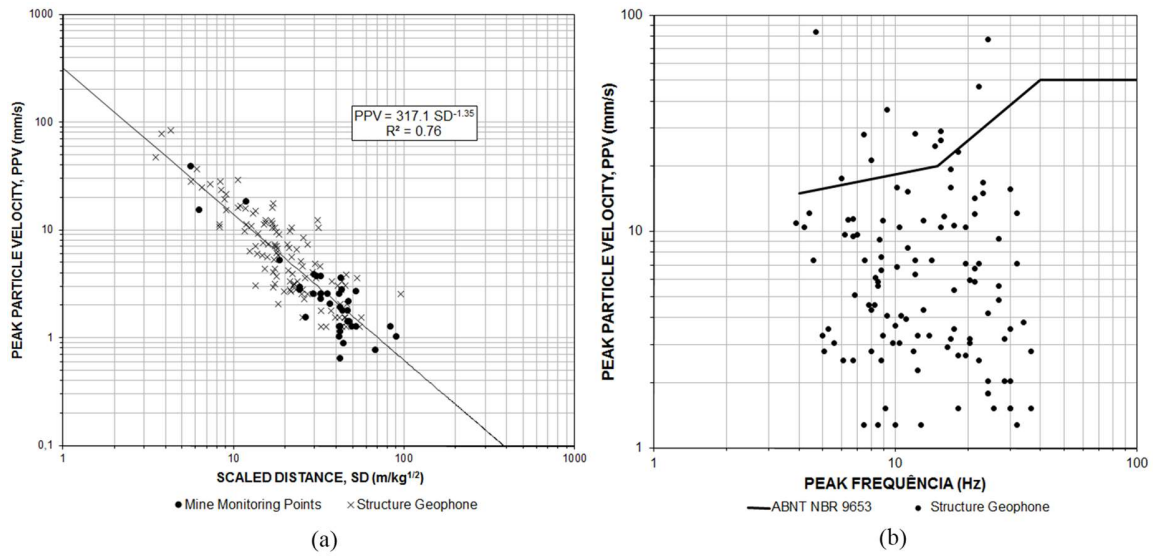


Fig. 6. Attenuation plot correlating peak particle velocity with scaled distance (a), peak particle velocity versus peak frequency the Brazilian standard limits [14] (b)

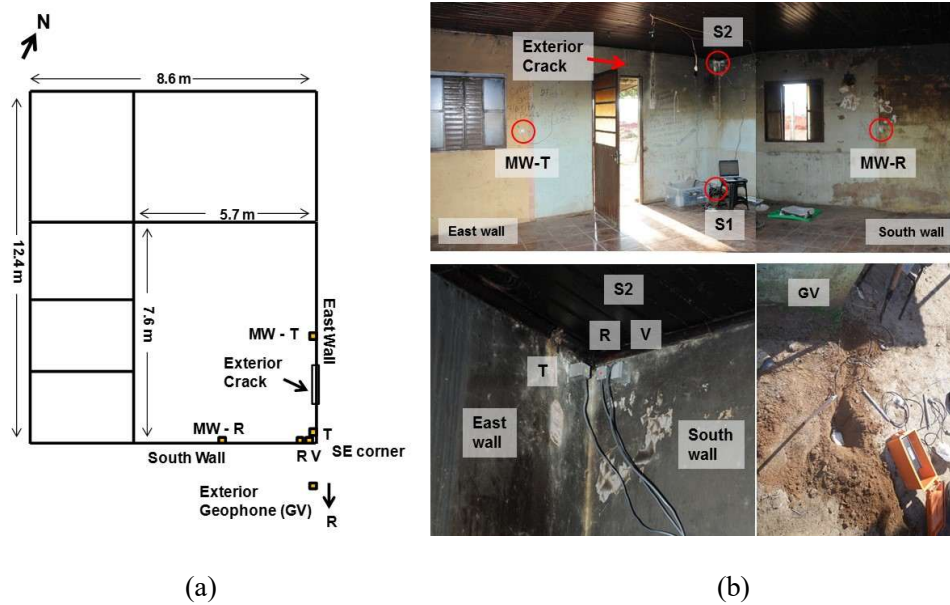


Fig. 7. Schematic drawing in plan view of the structure (a), photos of the instrumented walls and ground geophone installation (b)

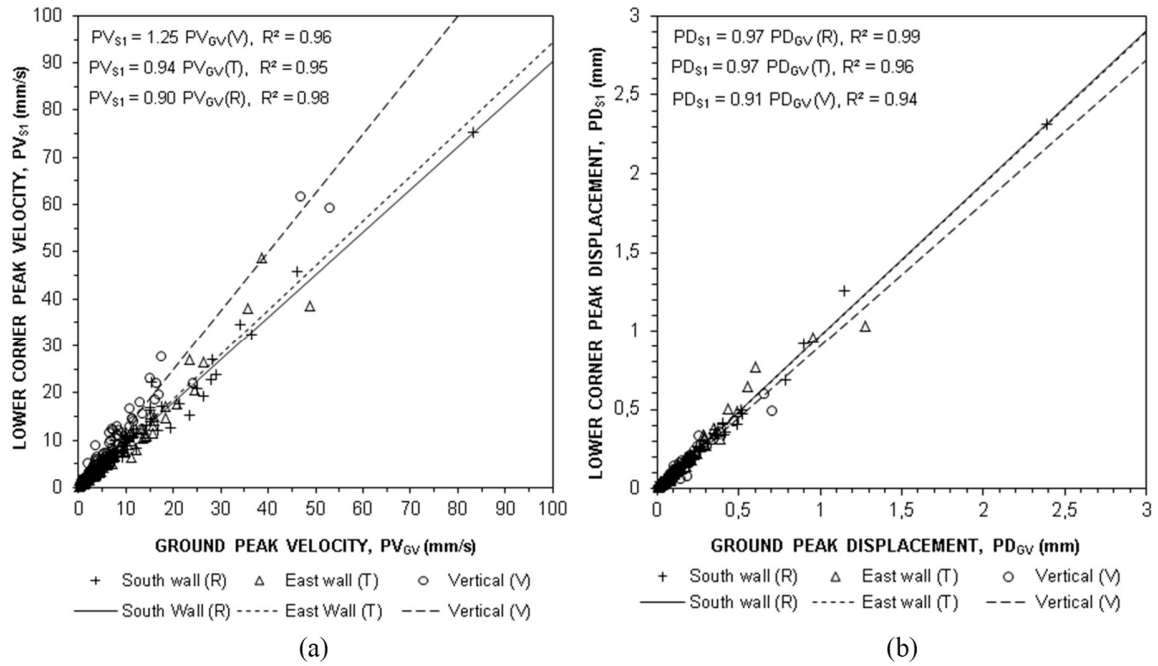


Fig. 8. Correlation of ground vibration (GV) with lower corner movement (S1), in terms of peak velocity (a) and peak displacement (b)

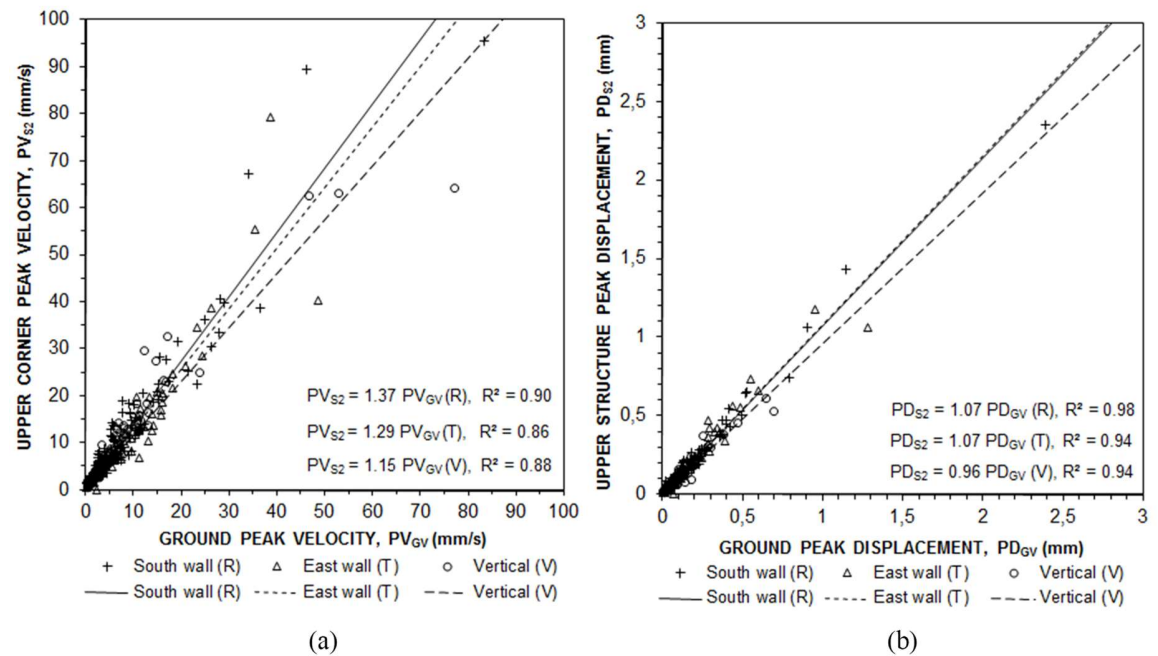


Fig. 9. Correlation of ground vibration (GV) with upper corner movement (S2), in terms of peak velocity (a) and peak displacement (b)

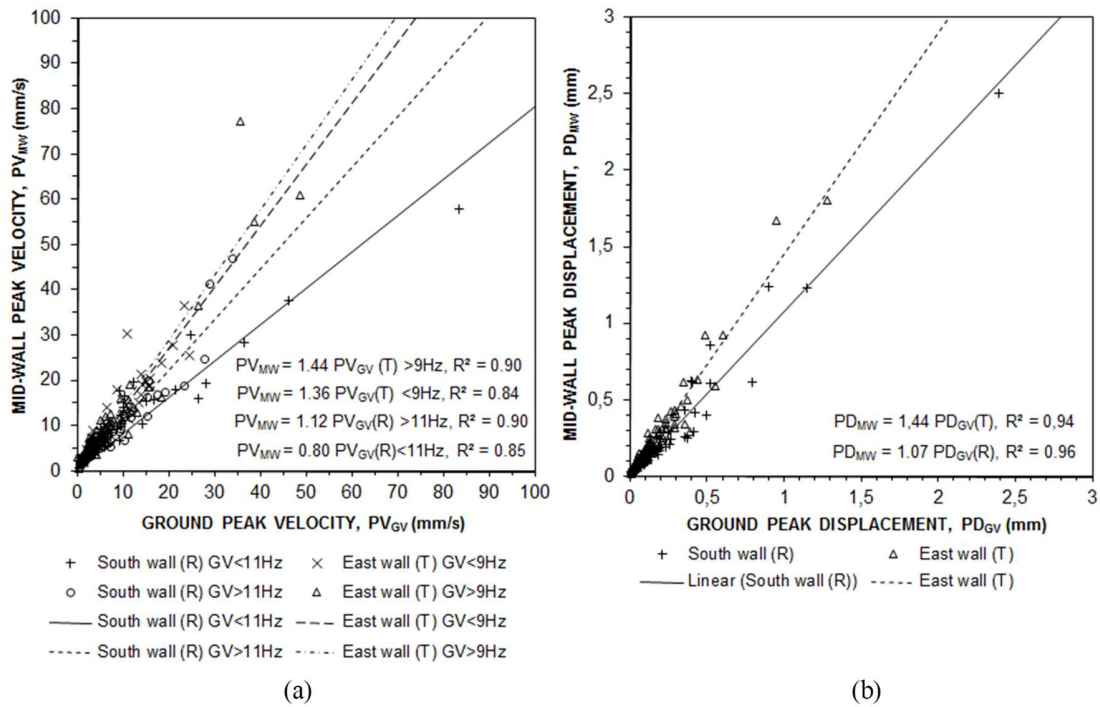


Fig. 10. Correlation of ground vibration (GV) with mid-wall movement (MW), in terms of peak velocity (a) and peak displacement (b)

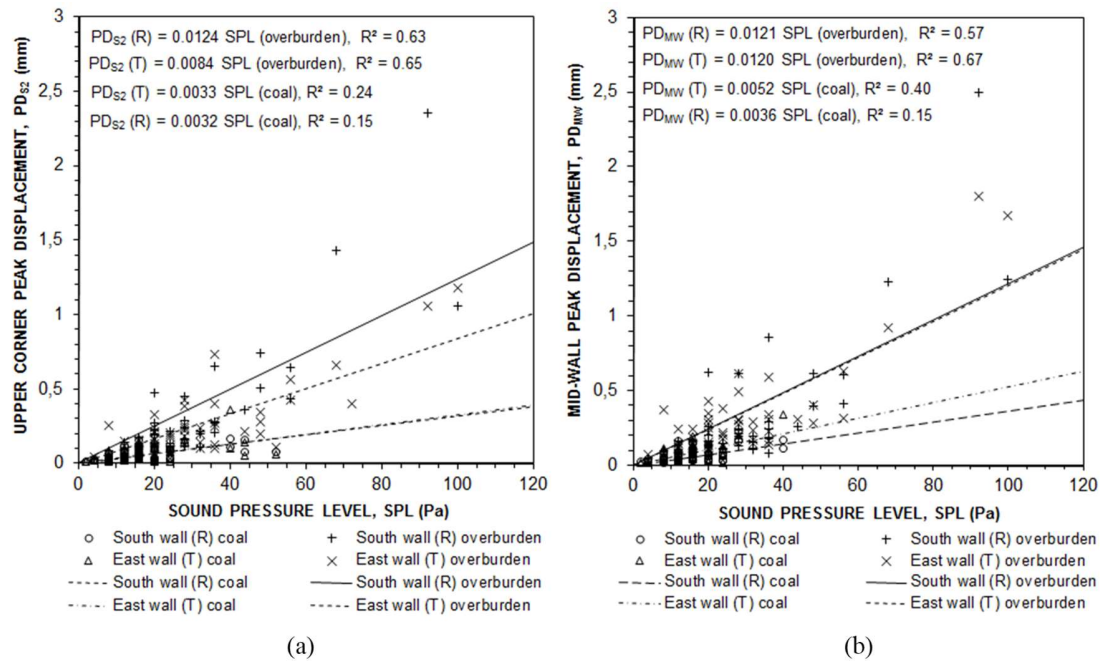


Fig. 11. Correlation of sound pressure level (SPL) with upper structure (a) and mid-wall (b) displacements

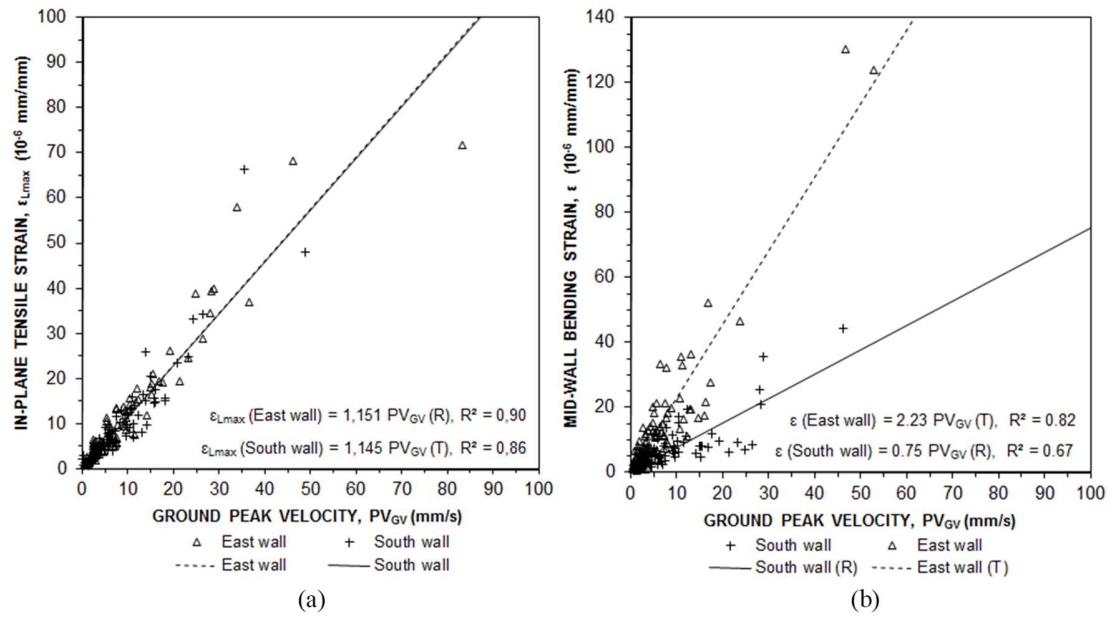


Fig. 12. Peak ground velocity influence on in-plane tensile strains (a) and bending strains (b)

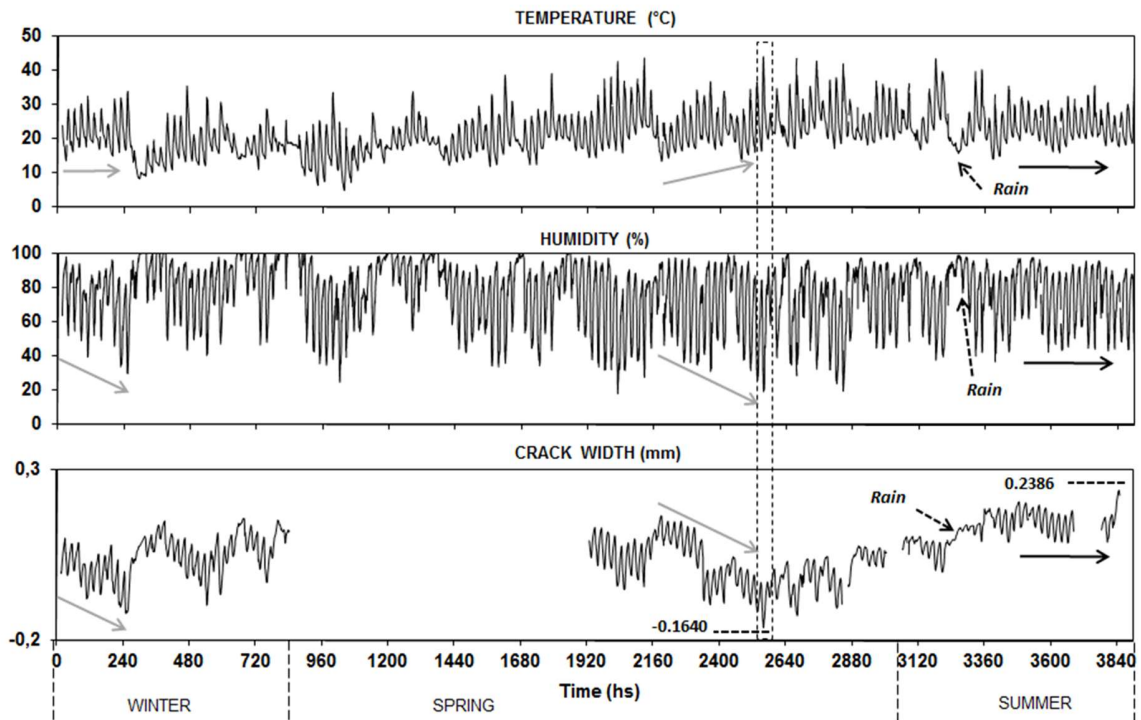


Fig. 13. Weather-induced daily changes in crack aperture

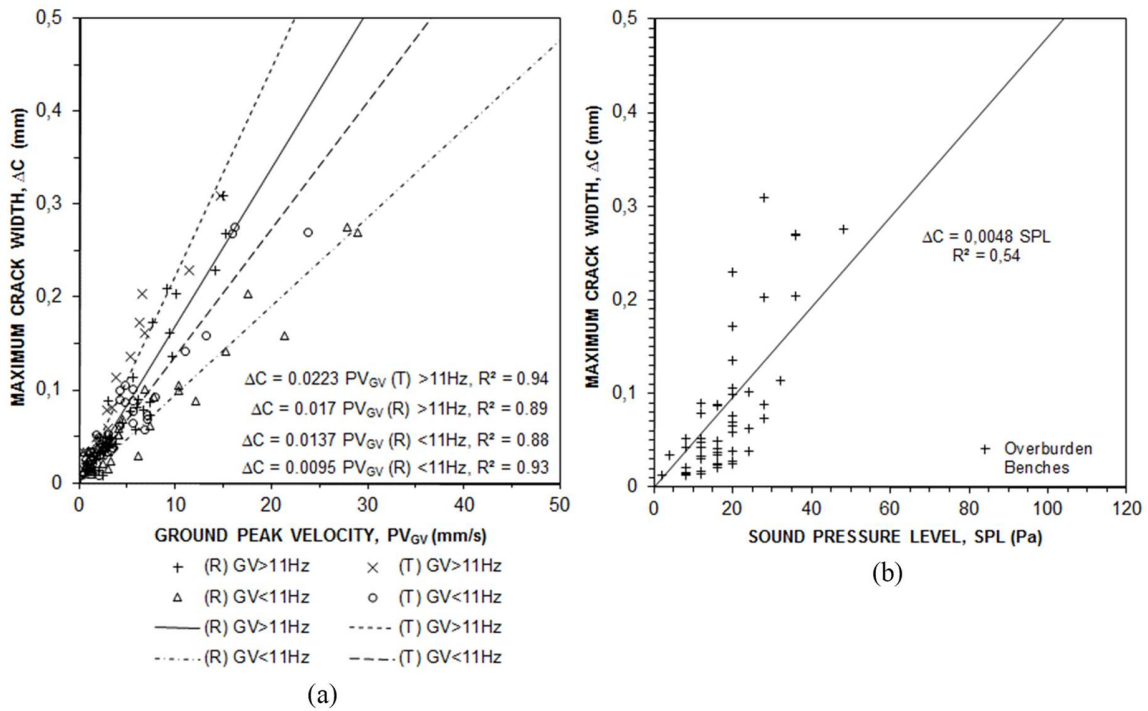
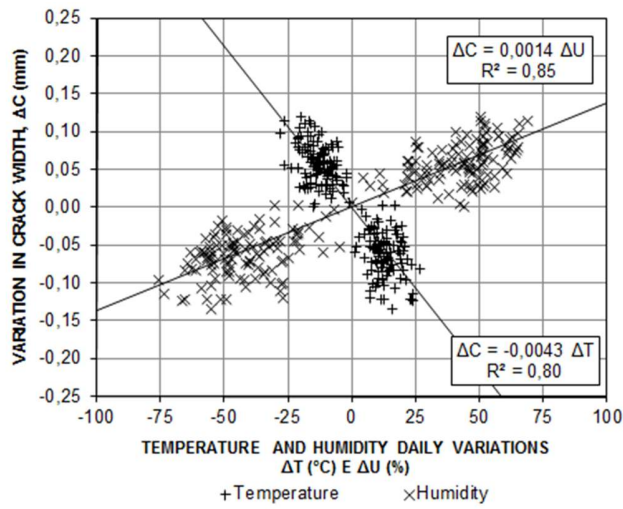


Fig. 15. Crack response to peak ground velocity (a) and sound pressure levels (b)

Table 1. Computed differential wall displacements, strain levels and safety factors against cracking for the highest blasts recorded at the structure

Peak Particle Velocity		Differential Displacement		Shear Strain		In-plane Tensile Strain		Mid-wall Bending Strain		Damage Potential	
R	T	South wall	East wall	South wall	East wall	South wall	East wall	South wall	East wall	In-plane Tensile Strain	Bending Strain
(mm/s)	(mm/s)	(mm)	(mm)	(micro-strains)	(micro-strains)	(micro-strains)	(micro-strains)	(micro-strains)	(micro-strains)		
28.19	24.38	0.333	0.226	90.4	133.0	33.3	39.5	20.9	27.7	3.8	5.4
26.42	15.75	0.243	0.098	39.2	97.2	14.4	28.8	8.6	13.4	5.2	11.2
36.58	13.97	0.312	0.102	40.8	124.8	15.0	37.0	2.5	19.2	4.0	7.8
17.53	13.21	0.162	0.056	22.3	64.7	8.2	19.2	11.9	18.2	7.8	8.3
27.94	20.83	0.291	0.160	63.8	116.4	23.5	34.6	25.3	21.6	4.3	5.9
21.34	14.99	0.163	0.140	56.0	65.0	20.6	19.3	6.3	36.3	7.3	4.1
46.23	38.61	0.575	0.319	127.6	230.0	47.0	68.3	44.4	61.1	2.2	2.5
24.89	23.37	0.328	0.168	67.1	131.0	24.7	38.9	6.8	52.1	3.9	2.9
28.96	26.42	0.337	0.230	93.2	134.8	34.3	40.0	35.6	46.5	3.7	3.2
83.31	48.77	0.603	0.326	130.4	241.2	48.0	71.6	17.6	123.9	2.1	1.2
34.04	35.56	0.489	0.450	180.0	195.6	66.3	58.1	87.2	130.5	2.3	1.1